

EDUCATION

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| <b>McGill University &amp; Mila - Quebec AI Institute</b><br><i>Master's of Science in Computer Science</i>                                                                                                                                                                                   | Montréal, QC<br><i>Aug. 2025 – Present</i>                |
| <b>Duke University / Duke Kunshan University</b><br><i>Bachelor's of Science in Data Science</i> <ul style="list-style-type: none"><li>Thesis: <i>Monitoring Spartina Alterniflora Using Self-Supervised.</i></li><li>Study Abroad: Universidad Carlos III de Madrid - Spring 2022.</li></ul> | Durham, NC / Suzhou, China<br><i>Aug. 2021 – May 2025</i> |

GRANTS & AWARDS

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| McGill CS Graduate Award — \$30,000/year               | 2025 |
| McGill CS Graduate Registration Bonus — \$5,000        | 2025 |
| DKU Student Experiential Learning Fellowship — \$1,000 | 2024 |
| DKU Signature Work Research Grant — \$1,500            | 2023 |
| DKU Innovation Incubator — \$3,000                     | 2023 |

RESEARCH EXPERIENCE

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| <b>Machine Learning Research Intern</b><br><i>University Mohammed VI Polytechnic</i> <ul style="list-style-type: none"><li>Unified a 10-yr global crop-yield to a fertilizer dataset (N/P/K) across 150+ countries.</li><li>Shipped transformer field-recs and an agentic LLM + retrieval copilot (RAG) with 2-second lookup latency.</li></ul>                                                                                                                                                                                                                                  | Jul. 2025 – Present<br><i>Benguerir, Morocco</i> |
| <b>Undergraduate Research Assistant</b><br><i>Duke Kunshan University</i> <ul style="list-style-type: none"><li>Built a 300k-image, ground-truthed satellite dataset across 50/54 African countries; trained DINO (ViT backbone) model to 97% accuracy (<i>supervised by Dr. Ka Leung Lam</i>).</li><li><i>SpeedTransformer</i>: SOTA 96% Top-1 on Geolife transport-mode prediction (+3%); launched <i>CarbonClever</i>, a social app turning mobility insights into personal CO<sub>2</sub> reductions (<i>supervised by Dr. Charles Chang</i>).</li></ul>                     | Jun. 2024 – Aug 2025.<br><i>Suzhou, China</i>    |
| <b>Undergraduate Research Assistant</b><br><i>University of Missouri</i> <ul style="list-style-type: none"><li>Built a Perceiver IO-based foundation model and a VAE simulator for wildfire spread using satellite + climate data, capturing long-range spatiotemporal dynamics and enabling scenario analysis (<i>supervised by Dr. Jordan Malof</i>).</li></ul>                                                                                                                                                                                                                | Jan. 2025 – Jun. 2025<br><i>Columbia, MO</i>     |
| <b>Undergraduate Research Fellow</b><br><i>Duke University</i> <ul style="list-style-type: none"><li>Mapped invasive salt-marsh species from Sentinel-2/Landsat using ViTs; delivered landscape-scale maps presented at Duke Climate+ 2024 to inform coastal conservation policy (<i>Supervised by Dr. Wenhong Li &amp; Dr. Ding Ma</i>).</li><li>Built a geospatial database on saltwater intrusion sea-level rise (North American Coastal Plain); mined 1,000+ papers with BERT; shipped an interactive ArcGIS dashboard (<i>Supervised by Dr. Emily Bernhardt</i>).</li></ul> | Jun. 2023 – Aug. 2024<br><i>Durham, NC</i>       |

PROFESSIONAL EXPERIENCE

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| <b>Data Analyst Intern</b><br><i>Atos Morocco</i> <ul style="list-style-type: none"><li>Built real-time HR dashboards used by leadership; scaled deployment across all African branches to standardize decision-making.</li></ul>                          | Oct. 2022 – Nov. 2022<br><i>Rabat, Morocco</i>        |
| <b>Data Analyst Intern</b><br><i>XPerlean</i> <ul style="list-style-type: none"><li>Applied Faster R-CNN and YOLO for ceramic defect detection, raising accuracy 70%→85% and boosting production speed 10% while reducing quality-control costs.</li></ul> | Jul. 2022 – Aug. 2022<br><i>Saint-Quentin, France</i> |
| <b>Data Analyst Intern</b><br><i>Al Jazeera Media Institute</i> <ul style="list-style-type: none"><li>Scraped and processed 200k+ social media records via APIs to analyze user behavior and engagement for reporting.</li></ul>                           | Oct. 2021 – Dec. 2021<br><i>Doha, Qatar</i>           |

## PUBLICATIONS & PREPRINTS

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**O. Echchabi**, et al. *Tracking Progress Towards SDG 6 Using Satellite Imagery*. arXiv:2411.19093.  
Z. Yuandong\*, **O. Echchabi\***, et al. *SpeedTransformer: Predicting Human Mobility with Dense GPS and Transformers* (Under Review).  
**O. Echchabi**, N. Talty. *Football AI Tracker: End-to-End Player Tracking from Low-Quality Recording*.

## SELECTED PRESENTATIONS

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Monitoring *Spartina alterniflora* Using Self-Supervised Learning. **Duke Climate+ Symposium, 2024**.  
Assessing Climate Change Risk of Rural Coastal Plains. **Duke Climate+ Symposium, 2023**.

## ACTIVITIES & VOLUNTEER WORK

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**Duke Kunshan University**: Resident Assistant; Student Athlete & Soccer Varsity Co-Captain; CS Club Software Developer [website]; Math & CS Teaching Assistant  
**FIFA World Cup 2022**: Team Leader, Spectator Services Volunteer  
**FADI Academy**: Partner & Math Tutor

## TECHNICAL SKILLS

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**Languages**: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R  
**Frameworks**: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI  
**Developer Tools**: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm  
**Libraries**: pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-Learn

## LANGUAGES

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Arabic (native); French (fluent); English (fluent); Chinese (intermediate); Spanish (conversational)